From Printed Critical Apparatus to Structured Data: A Case Study on the Benedictine Vulgate

Anonymous Submission

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Abstract

Critical editions contain enormous quantities of expert-curated data on textual variation, but much of it remains inaccessible to computational analysis because it is encoded in highly compressed printed *critical apparatuses*. This paper presents a four-stage pipeline combining a Vision-Language Model, a Convolutional Neural Network classifier, a rule-based adjudication stage, and a formal-language parser to transform a printed apparatus into a structured dataset. Our use case is the apparatus to the Book of Genesis in the Benedictine edition of the Vulgate, a work historically associated with the development of quantitative approaches to stemmatology. On a manually annotated ground truth comprising the first four chapters of Genesis, the pipeline achieves a character error rate of 2.11% and an F_1 score of 92.7% for witness extraction. Additionally, we demonstrate that the automatically extracted data closely reproduce the witness-agreement patterns observed in the ground truth, suggesting that the pipeline preserves much of the structure relevant for downstream quantitative analysis. Finally, we release the first structured dataset covering the entire apparatus of the book of Genesis, as a JSON file and as a web application designed for easy visualization and exploration of this landmark philological work.

Keywords: Document parsing, Textual criticism, Computational philology

1 Introduction

Critical editions contain enormous quantities of expert-curated data on textual variation, but much of it remains inaccessible to computational approaches because it is encoded in highly compressed printed *critical apparatuses*, *i.e.* systems of notes recording variant readings across the manuscript witnesses used to establish the text. Although recent advances in *Optical Character Recognition* (OCR) have substantially improved the transcription of printed and handwritten documents, critical apparatuses remain largely unexplored data, leaving valuable information locked in print and unavailable for computational textual criticism.

A compelling example is the *Biblia Sacra iuxta Latinam Vulgatam Versionem ad codicum fidem*, or Benedictine Vulgate, a major critical edition of the Old Testament in its Latin Vulgate version. Its first editor, the philologist Dom Henri Quentin, developed one of the earliest systematic quantitative methods for textual reconstruction, usually referred to as the "triplet method" (*méthode des triplets*), based on patterns of agreement among witnesses [26]. He used this method to classify the witnesses of the Octateuch into textual families and propose their possible interrelationships [27]. Criticized both conceptually and for the amount of manual computation it required, the method was seldom applied

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thereafter [2; 28], and Quentin's analysis has consequently never been reassessed using modern methods for manuscript classification and stemmatology [10; 11; 13; 14; 21; 24]. While revisiting these results through the prism of modern computational tools could yield very interesting insights for the study of the transmission of the Latin Bible, such an undertaking runs into the problem of data availability: no machine-readable dataset, whether of raw transcriptions or of collations, exists for the Latin witnesses Quentin used, and redoing the transcriptional work from scratch would be prohibitively expensive. An alternative approach would be to rely on the edition's critical apparatus, which already contains the required information, but it must first be digitized before any computational analysis can be performed.

We therefore propose a modular four-stage pipeline that converts a densely encoded critical apparatus into structured data. It combines an off-the-shelf *Vision-Language Model* (VLM), a classifier trained on synthetic data to correct edition-specific glyphs, a reconciliation stage integrating both outputs, and a context-free grammar that parses the resulting transcription into structured JSON. We apply this pipeline to the apparatus of Genesis in the first volume of the Benedictine Vulgate [25], producing the first machine-readable dataset for a corpus that was itself among the earliest objects of quantitative textual criticism.

Evaluation against a manually established ground truth for Genesis 1–4 yields a 2.11% *Character Error Rate* (CER) and an F_1 score of 92.7% for witness extraction, compared with 61.7% and 47.8% respectively when relying on the off-the-shelf VLM alone. Additionally, the witness partitions extracted using a naive clustering approach on the pipeline data remain highly consistent with the ground truth, with *Adjusted Rand Index* (ARI) values of 0.75–1.00 across the evaluated clustering resolutions. This consistency is also reflected in the pairwise witness-agreement counts, which correlate strongly between the pipeline and ground-truth data (Pearson's r=0.991; Spearman's ρ=0.931). We release the complete Genesis apparatus as a silver-standard JSON dataset, together with a companion website.[1]

The contributions of this study are thus threefold:

- A modular pipeline architecture that decomposes apparatus digitization into four stages, instantiated and evaluated on the book of Genesis of the Benedictine Vulgate;
- An end-to-end evaluation on a manually established ground truth for Genesis 1–4;
- The release of the complete Genesis apparatus as a silver-standard structured dataset, together with a companion website.

The paper is structured as follows. Section 2 presents the specificities of the Benedictine Vulgate's critical apparatus and shows how it can be understood as a formal grammar. Section 3 reviews the current state of the art. Section 4 presents our pipeline, Section 5 presents our evaluation methodology, and Section 6 presents the pipeline's performance. Section 7 is dedicated to the identified limits of this study, and Section 8 presents our conclusions and future work.

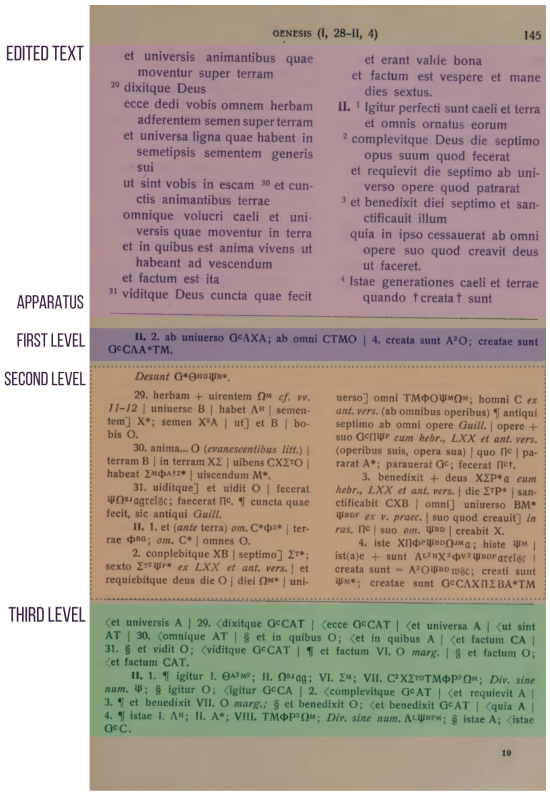
[1] During anonymous review, a screenshot of the website is provided in Figure 10 and the public link will be added after review.

2 Derivation rules of the critical apparatus of the Benedictine Vulgate

A critical apparatus records the variant readings attested by the witnesses to a text according to an edition-specific syntax, as documented in its preface.[2] The Benedictine Vulgate uses three apparatuses: the first reports readings from the principal codices, the second variants from all examined codices, and the third differences in textual division. Because our objective is to recover agreements among the complete set of witnesses, we process only the second-level apparatus, as shown in Figure 1a.

Figure 1: Structure and notation of the Benedictine Vulgate apparatus.

(a) Page showing the three apparatus levels; this study processes the second.



(b) Apparatus for Gen. 50:24 and its interpretation.

24. iurasset B*Φᴬ* | adque GCXΠᶜ Θᴴ | uisitauit GΛᴸΠᶜΣᵀ*ᴹBAT*P* | portate Π | uobiscum *om.* Πᴰ | ∼ ossa mea uobiscum XA agrelʊgc.

No.	Type	Reading and witnesses
1	substitution	*iurasset*: B^*, Φ^A
2	substitution	*adque*: G, C, X, Π^C, Θ^H
3	substitution	*uisitauit*: $G, \Lambda^L, \Pi^C, \Sigma^{T*}, \Sigma^M, B, A, T^*, P^*$
4	substitution	*portate*: all witnesses with a siglum with base Π
5	omission	*uobiscum*: Π^D
6	transposition	*ossa mea uobiscum*: X, A, and the printed editions agrelʊsc

The apparatus is organized by chapter and verse. The entries for each verse contain *units*, separated by |, corresponding to variant loci; units may in turn contain several *segments*, separated by ;, each associating a reading with its supporting witnesses. Critical *signs*, *i.e.* conventional symbols encoding the relationship between the lemma contained in the established text of the edition and the variants found in the manuscripts, indicate the variant readings and their type. In the case of the Benedictine Vulgate, five types of information are distinguished: substitution, indicated by] when the lemma is explicit or otherwise by the reading of the variant alone; omission (*om.*); addition (+); transposition (∼); and positive apparatus (=), which explicitly lists the witnesses supporting the edited reading. By the editor's own admission, the edition generally employs a negative apparatus, listing only witnesses that diverge from the edited text, and uses the positive form

[2] For the Benedictine Vulgate, see *Mémoire sur l'Établissement du Texte de la Vulgate*[27], pp. 510–13, for a full description of the editorial choices regarding apparatus notation.

irregularly. Paratextual information, such as marginal content, is introduced by ¶.
Witnesses are identified by their *sigla*, consisting of Latin or Greek base letters, optionally with superscripts. Sigla sharing the same base are contracted; for example, Φ^{AR} represents Φ^{A} and Φ^{R}. Superscript symbols further distinguish manuscript states: † marks a first-hand reading altered by the original scribe, whereas * marks a first-hand reading corrected by another hand; ^{1}, ^{2}, and ^{3} identify successive correctors. Thus, $\Phi^{AR^{1}}$ denotes a reading attested in the witness Φ^{A}, as well as found in Φ^{R} as an addition made by the first correcting hand. These densely packed and visually similar symbols pose a major challenge for off-the-shelf OCR, as will be demonstrated in the results section.
Segments may also record agreement with ancient versions or patristic citations, introduced by *cum* and *cf.*, and may include readings from medieval correctoria. Editorial annotations sometimes conclude a unit. An example of the apparatus entry of Gen 50:24 and its interpretation is available in Figure 1b.
These strictly structured conventions allow the second-level apparatus to be modeled as a context-free grammar, as expressed in extended Backus–Naur form (EBNF) notation in Figure 2. The grammar captures the hierarchical organization and editorial syntax of the apparatus, while free-text readings, citations, and annotations are represented as terminal fields whose internal linguistic structure is not further parsed, as considered beyond the scope of the current study, which focuses on witness agreement. Figure 3 illustrates the derivation of an excerpt from the apparatus entry for Gen. 4:24.

3 Related work

Recent advances in document understanding and vision-language models (VLMs) have considerably improved the extraction of structured information from printed documents [9; 12; 35]. Recent work in document understanding has moved beyond flat character transcription toward direct extraction of structured information from page images, using end-to-end vision transformers to output structured markup from documents [6; 16], such as invoices, financial statements, or contracts [17; 18]. Their application to critical apparatuses, however, remains largely unexplored, as previous research on critical editions has primarily focused on recognizing the main text and its reference structure [1] rather than on converting the apparatus into structured data.
To our knowledge, the work of Bambaci et al. provides the only previous attempt to convert a critical apparatus automatically into structured data. Developed within the *DiKe–TexTHBM* and *Reversing Kennicott* projects [5], their approach targets the apparatus of Benjamin Kennicott's *Vetus Testamentum Hebraicum cum variis lectionibus* [15]. Like our suggested approach, they model the transcribed apparatus as a domain-specific language and parse it through rule-based procedures [3; 4].
Kennicott's and the Benedictine Vulgate editions nevertheless pose different recognition problems, as Kennicott's apparatus primarily combines Hebrew readings with manuscript identifiers expressed as Arabic numerals, whereas the Benedictine Vulgate uses Latin readings alongside a dense system of manuscript sigla, typographical modifiers, and editorial signs. Since modern OCR systems generally handle Latin readings adequately, the main difficulty in our case study lies in recovering witness lists and interpreting the information encoded in their typography. Figure 4 illustrates a particularly challenging example.
This typographical complexity consequently required us to supplement an off-the-shelf VLM with a dedicated glyph classifier with a lightweight convolutional neural network, specifically trained to recognize manuscript sigla and editorial signs, a step not needed in Bambaci's study. Such an approach, combining a general-purpose OCR system with a specialized recognizer, is an established strategy in OCR research [23] and has previously

Figure 2: Production rules of the apparatus.

```
apparatus   → verse { newline verse } ;
verse       → verse_num "." unit { "|" unit } ;
unit        → [ word { word } "]" ] segment { ";" segment } ;
segment     → [ "¶" ] content witness_group [ trailer ] ;
content     → omission | addition | positive | transpo | sub ;
omission    → { word } "om." ;
addition    → word { word } "+" word { word } ;
transpo     → "~" word { word } ;
positive    → word { word } "=" ;
sub         → word { word } ;
witness
_group      → witness_subgroup { witness_subgroup } ;
witness
_subgroup   → manuscript_group | edition_group ;
manuscript
_group      → siglum { siglum } [ "in ras." ] ;
edition
_group      → edition { edition } ;
siglum      → base [ sup_arg ] ;
sup_arg     → "^" ( hand | "{" sup "}" ) ;
sup         → sup_item { sup_item } ;
sup_item    → letter [ sup_arg ] | hand ;
base        → Θ | Φ | Ψ | Ω | Λ | Σ | Π
              | A | B | C | G | M | O | P | T | X ;
letter      → A | B | C | D | F | G | H | J
              | L | M | O | P | S | T | V | Z ;
hand        → "*" | 1 | 2 | 3 | † ;
edition     → 𝔞 | 𝔠 | 𝔢 | 𝔤 | 𝔩 | 𝔯 | 𝔰 | 𝔳 | 𝔴 ;
trailer     → ( "cum" | "cf." ) ( version_list | citation_list )
              | annotation ;
version
_list       → version { ( "et" | "," ) version } ;
citation
_list       → citation { ";" citation } ;
```

been applied to recognize rare or domain-specific vocabularies [34]. Previous research has also shown that outputs from multiple OCR systems can be combined through sequence alignment and voting procedures to determine which output to select [19; 20].
Our pipeline adapts this strategy to the constrained syntax of a critical apparatus. The choice of this modular design was motivated by the need to avoid end-to-end fine-tuning of a VLM, as it would first require manually producing a substantial structured training dataset, the very annotation effort that the proposed pipeline is intended to avoid.

4 Parsing pipeline

4.1 Global overview of the pipeline

Our pipeline is thus organized into four main stages, as schematized in Figure 5:

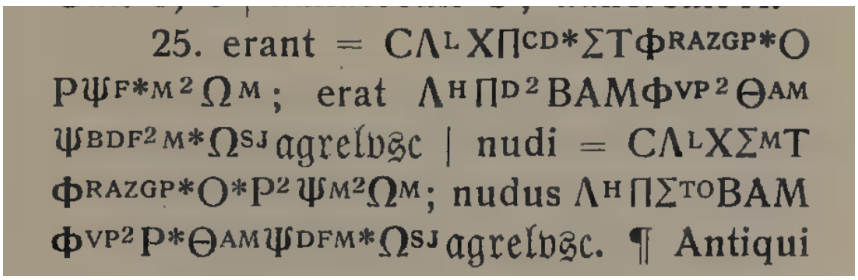
1. An off-the-shelf VLM processes each page to produce an initial transcription and detect the page layout.

Figure 3: Example of derivation of one verse entry (Gen. 4:24). For brevity, the first segment of unit 3 is not fully derived as it follows the same pattern as unit 1, and only one siglum is fully derived, as all sigla follow the same derivation pattern.

Input (Excerpt of Gen. 4:24).
`24. septumplum` G^*B `| ulcio + enim` Ω^{S^*} `| dauitur` A`; om.` O

(1)	*apparatus*	$\Rightarrow$ *verse*
(2)	*verse*	$\Rightarrow$ `verse_num .` *unit*$_1$ `|` *unit*$_2$ `|` *unit*$_3$
(3)		$\Rightarrow$ `24 .` *unit*$_1$ `|` *unit*$_2$ `|` *unit*$_3$
(4)	*unit*$_1$	$\Rightarrow$ *segment*$_1$
(5)	*segment*$_1$	$\Rightarrow$ *content witness_group*$_1$
(6)		$\Rightarrow$ *sub witness_group*$_1$
(7)		$\Rightarrow^*$ `septumplum` *witness_group*$_1$
(8)	*witness_group*$_1$	$\Rightarrow$ *witness_subgroup*
(9)		$\Rightarrow$ *manuscript_group*
(10)		$\Rightarrow^*$ G^*B
(11)	*unit*$_2$	$\Rightarrow$ *segment*$_2$
(12)	*segment*$_2$	$\Rightarrow$ *content witness_group*$_2$
(13)		$\Rightarrow$ *addition witness_group*$_2$
(14)		$\Rightarrow^*$ `ulcio + enim` *witness_group*$_2$
(15)		$\Rightarrow^*$ `ulcio + enim` Ω^{S^*}
(16)	*unit*$_3$	$\Rightarrow$ *segment*$_3$ `;` *segment*$_4$
(17)	*segment*$_3$	$\Rightarrow^*$ `dauitur` A
(18)	*segment*$_4$	$\Rightarrow$ *content witness_group*$_4$
(19)		$\Rightarrow$ *omission witness_group*$_4$
(20)		$\Rightarrow^*$ `om.` O
(21)	*siglum*	$\Rightarrow$ *base sup_arg*
(22)		$\Rightarrow$ *base* `^` *hand*
(23)		$\Rightarrow^*$ *base* `^*`
(24)		$\Rightarrow^*$ G^*

Figure 4: Example of a complex witness list requiring transcription and parsing, taken from the apparatus of Gen 2:25.



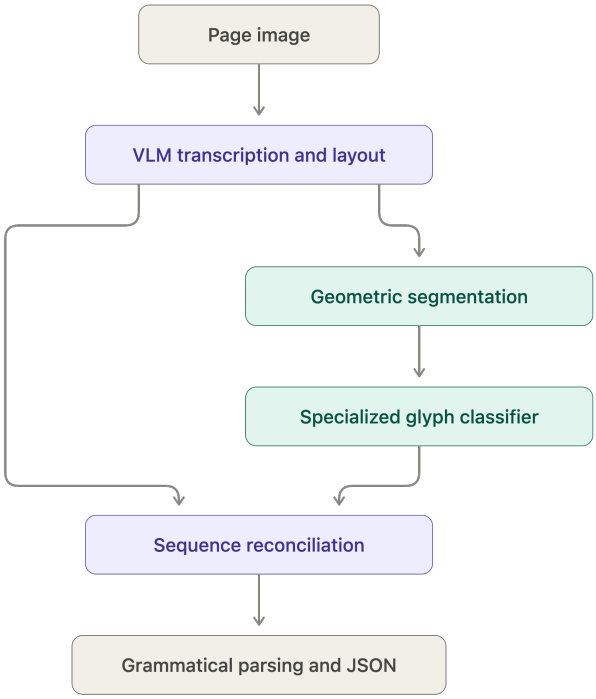
2. Using the detected layout and page geometry, the apparatus is localized, deskewed, and segmented into lines and token crops. Each crop is then analyzed by a lightweight

convolutional neural network trained to recognize the manuscript sigla, editorial signs, and special typographical forms used in the edition.

3. The VLM spans and classifier tokens are aligned through global sequence alignment and reconciled using the classifier confidence, local siglum-formation constraints, and geometry-based rules.

4. A deterministic parser based on the grammar described in Section 2 identifies readings, witnesses, variant types, and contextual information, and serializes them as structured JSON.

Figure 5: Overview of the transcription and parsing pipeline



4.2 Transcribing the apparatus

4.2.1 VLM transcription and image segmentation

The first stage applies an off-the-shelf VLM to each page. After preliminary experiments with several open-weight models (CHURRO [31], DeepSeek-OCR [35], and Chandra-OCR [7]), we selected GLM-OCR [8; 9] as we found it provided the best trade-off between transcription quality and computational efficiency on our material with preliminary experiments on pages outside the evaluation corpus. GLM-OCR combines the PP-DocLayoutV3 layout detector [33] with a lightweight recognition model, producing a transcription and bounding boxes for the detected page regions.

The second-level apparatus is then localized from the VLM bounding boxes using the horizontal rules. The resulting crops are then deskewed and segmented into lines using horizontal projection profiles. Individual glyphs are then detected by grouping connected

pixels, and small elevated marks are attached to the character below them to reconstruct sigla containing superscripts. Each resulting token and its constituent components are normalized to 64×64-pixel crops for classification.

4.2.2 Specialized glyph classifier

Each segmented token crop is then analyzed by a lightweight multi-head convolutional neural network trained on the edition's typography. Its recognition inventory, derived from the witness and sign lists in the preface, comprises manuscript sigla, Fraktur characters, editorial symbols, and a background class that prevents ordinary text and punctuation from being interpreted as specialized symbols.

A shared convolutional network uses separate output heads to classify the token, its base character, and any modifiers. For composite tokens, an additional head classifies the individual image components, which are read from left to right to reconstruct sequences of superscripts or Fraktur characters.

Because no existing dataset represents this combination of typeface, sigla, and editorial conventions, we created a dedicated training dataset from 215 reference crops collected from the preface and apparatus pages, outside of the evaluation chapters (Genesis 1–4). These forms were decomposed into reusable bases and modifiers, then recombined according to the edition's inventory of valid sigla. To approximate their appearance in the digitized volume, the synthetic glyphs were resized to the 20–24 pixel cap height measured on the 330-DPI scans and composited onto ink-free paper patches sampled from the same pages. Augmentation reproduced plausible printing and digitization defects through small rotations, stroke dilation and erosion, broken strokes, variations in ink darkness, brightness and contrast jitter, Gaussian blur, and JPEG compression. Examples of the resulting training images are shown in Figure 6.

Figure 6: Examples of data used to train the specialized glyph classifier.



(a) Forms extracted from the edition.



(b) Composition of synthetic sigla.



(c) Simulated degradation.

The synthetic data were then supplemented with the manual crops taken from the apparatus pages, for a total of 1157 manuscript tokens, 977 isolated modifiers, and 520 Fraktur characters. Crops from the same page were assigned to a single partition to prevent page-level overlap. Ordinary text, punctuation, and recurrent false detections were also included as negative examples. The final dataset comprised 30703 samples.

The prediction heads were trained jointly for 40 epochs using a cross-entropy loss. For composite sigla, a successful match required correct recovery of the base, all superscripts, and any numeral. The resulting model contains 0.79 million parameters. On a held-out 10% test partition, it achieved exact-match accuracies of 95.8% for composite manuscript sigla, 99.4% for Fraktur letters, and 100% for the comparatively simple editorial signs indicating variant types.

4.2.3 Reconciling the VLM and classifier outputs

We design this reconciliation procedure as an edition-specific error-correction layer targeting three recurrent failure modes observed on the development pages: modifier confusion within otherwise correctly recognized sigla, alignment shifts among repeated sigla, and spurious detections of Fraktur characters or editorial signs. As the two recognition streams

provide complementary information (VLM the running text, the classifier the specialized editorial symbols), we reconcile them asymmetrically rather than treating them as equivalent OCR outputs.
Sigla, *Fraktur* sequences, and editorial signs are first extracted from the VLM transcription and aligned with the classifier output using the Needleman–Wunsch algorithm [22] with a custom cost function. The custom alignment cost function prioritizes exact matches and uses the classifier's confidence to resolve disagreements and unmatched detections.
Aligned pairs are then adjudicated using classifier confidence, local siglum-formation constraints, and approximate image position. When both streams agree on a siglum's base but disagree on its modifiers, the VLM reading is retained by default. The classifier output replaces it only when: (i) both detections occupy the same image region and modifier confidence is above 0.9; (ii) the classifier identifies with high confidence a visually similar modifier using a manually compiled list, such as superscript Z where the VLM output contains 2; or (iii) the VLM proposes a modifier combination absent from the edition's inventory of valid sigla.
Fraktur pairs are retained only when the two readings substantially overlap, either through subsequence agreement or agreement on at least half of their characters. This prevents false-positive *Fraktur* detections from replacing unrelated lowercase text. A fuller description of a reconciliation algorithm is available in Appendix B and the implementation is available on the companion GitHub. All rules and thresholds were fixed on pages outside the evaluation corpus and applied unchanged during evaluation.

4.3 From transcription to structured data

The reconciled transcription is then converted into structured JSON using a deterministic parser implementing the formal grammar introduced in Section 2. Unlike the local siglum-formation constraints used during reconciliation, this grammar models the hierarchical relationships among verses, units, segments, readings, and witness groups.
This transformation is performed in three stages: grammatical parsing, expansion and normalization of witness identifiers, and semantic classification of readings. Before parsing, typographically equivalent Unicode and LaTeX encodings of Greek letters, superscripts, *Fraktur* characters, and editorial signs are mapped to a canonical representation. Layout-dependent elements, such as line breaks and page markers, are removed, while chapter and verse boundaries are retained as explicit structural markers.

4.3.1 Grammatical parsing

The normalized transcription is parsed according to the grammar defined in Section 2. The parser first divides the input into verses and apparatus units, then decomposes each unit into its constituent segments. Within each segment, it identifies the reported reading, associated witness groups, any critical sign, and optional trailing information.
Lexical categories are recognized from the most to the least constrained. Elements belonging to closed inventories, *i.e.*, witness bases, superscripts, edition sigla, critical signs, and abbreviations for ancient versions, are identified before generic textual tokens. This ordering limits ambiguity between apparatus notation and the words constituting the reported readings. When a verse cannot be fully parsed according to the grammar, the parser emits a partial record containing the successfully parsed segments and flags the verse for manual review in a separate logging file.

4.3.2 Normalizing witness identifiers

Witness expressions recognized by the grammar are converted into atomic identifiers suitable for quantitative comparison. Contracted sigla are then expanded according to the finite inventory described in the preface. For example, Φ^{RA^2ZGP} is expanded into Φ^{R}, Φ^{A^2}, Φ^{Z}, Φ^{G}, and Φ^{P}. Hand and state modifiers remain attached to the witnesses to which they apply. Stand-alone manuscript and edition sigla are likewise stored. Witnesses that cannot be reconciled with allowed sigla are, in practice, marked as such to speed up manual correction. To keep our evaluation protocol fully automatic, the parser neither corrects nor discards identifiers absent from the edition's inventory.

The parser distinguishes direct manuscript witnesses from ancient versions and patristic or editorial references. Evidence from other textual traditions is normalized to a closed vocabulary comprising Hebrew, *LXX* for the Septuagint, and *OL* for the Old Latin. These categories are stored separately from manuscript witnesses, allowing their later use in analyses of interlingual relationships without conflating direct and indirect evidence.

Because the present study focuses on agreement among direct witnesses, patristic citations and references to medieval critical sources are retained as annotations but are not further decomposed into author or work. This conservative treatment preserves the information without introducing potentially uncertain semantic interpretation of notes expressed in natural language.

4.3.3 Classifying variant types

Once a segment has been parsed, deterministic rules classify its variant type from the critical sign and its syntactic position. Each resulting record therefore associates a variant operation with its textual scope, reading, and supporting witnesses. When no explicit sign is present, this indicates substitution. Qualifications concerning the location or condition of a reading are retained in a separate annotation field for further post-processing. Readings present in the margin are stored with a location attribute.

4.3.4 Contextual information and final output

The parser also maintains two forms of document-level state: the current chapter and declarations of unavailable witnesses. The latter distinguish between witnesses absent due to lacunae and those omitted because they support the edited text. Since these declarations are made on a per-page basis, they are kept while the corresponding page is parsed and are updated when a new declaration is encountered.

The resulting records are serialized as JSON and grouped by chapter and verse. Figure 7, corresponding to the derivation shown in Figure 3, illustrates the output structure. Each record contains the variant type, the reading printed in the edition where applicable, the alternative reading, and its supporting witnesses. Optional fields record evidence from ancient versions, citations, editorial annotations, marginal information, and unavailable witnesses.

To assess the parser's performance independently of OCR performance, we applied the parser to the manually established ground-truth transcription and manually verified the resulting JSON against the source. No parsing or normalization errors were found in this evaluation corpus under error-free transcription. The end-to-end evaluation therefore compares the structured output obtained from the pipeline transcription with this verified structured ground truth.

Figure 7: Excerpt of the critical apparatus of verse 24, chapter 4 of Genesis structured as a JSON file, corresponding to the derivation in Figure 3. Note that the names of the manuscripts in the final dataset are encoded according to LaTeX conventions.


```
{
  "chapter": "4",
  "verse": "24",
  "missing": [
    "Θ^H", "Θ^G", "Ψ^B"
  ],
  "apparatus": [
    {
      "variant_type": "substitution",
      "alternative_reading": {
        "reading": "septumplum",
        "witnesses": ["G*", "B"]
      }
    },
    {
      "variant_type": "addition",
      "anchor": "ulcio",
      "alternative_reading": {
        "reading": "enim",
        "witnesses": ["Ω^S*"]
      }
    },
    {
      "variant_type": "substitution",
      "alternative_reading": {
        "reading": "dauitur",
        "witnesses": ["A"]
      }
    },
    {
      "variant_type": "omission",
      "alternative_reading": {
        "witnesses": ["O"]
      }
    }
  ]
}
```


5 Evaluation methodology

5.1 Aligning the pipeline output and the ground truth

We evaluate the pipeline against a fully hand-annotated ground truth covering the first four chapters of Genesis [25]. It comprises 25923 transcribed characters, 106 verses, and 936 apparatus entries, containing 2868 sigla and 8860 characters in textual fields. The evaluation corpus contains instances of all five variant types distinguished by the apparatus (substitution, omission, addition, transposition, and positive apparatus), all editorial signs, contracted and uncontracted sigla, multi-segment units, composite superscript groups, trailers with ancient versions and patristic citations, marginal readings introduced by ¶, and witness unavailability declarations (Desunt). The ground truth was produced by a single annotator copying the apparatus directly from the available pages, as the printed text presented no ambiguity.

The ground-truth transcription was then converted to JSON using the parser described

in Section 4.3, and, as noted above, we found no parsing or normalization errors in the resulting JSON. We thus compare the two structured datasets directly. To do so, the two JSON outputs are first aligned at the chapter-and-verse level. A chapter-verse pair present in the ground truth but absent from the pipeline output is marked as missing.
Within each chapter-verse pair, apparatus entries cannot simply be aligned by index, since a missing or spurious entry would shift all subsequent correspondences. We therefore align the predicted and ground-truth entry sequences using the Needleman–Wunsch algorithm [22].
Candidate pairs are scored by equally weighting normalized Levenshtein similarity across their textual fields and agreement in variant type. Witness-set similarity is deliberately excluded because witness recovery is itself an evaluated outcome. As a robustness check, we repeated the alignment under alternative weighting schemes, including one that incorporates witness-set similarity, and obtained identical correspondences. Pairs with a similarity score below 0.60 are treated as unmatched. We assessed the robustness of this threshold by inspecting the alignments obtained with thresholds of 0.20, 0.40, 0.60, and 0.80. Thresholds between 0.20 and 0.60 produced identical correspondences, and we retained 0.60 as the most conservative value that preserves all valid pairs. We manually inspected the non-matches to ensure they were due to OCR errors rather than the alignment process, and that they did not constitute plausible correspondences between entries representing the same apparatus locus.
The resulting one-to-one correspondences form the basis of the evaluation metrics reported below.

5.2 Selected evaluation metrics

As the purpose of the pipeline is not merely to transcribe printed pages, but to recover apparatus data in a structured representation suitable for textual criticism, we evaluate its performance along four dimensions: (**i**) apparatus structure recovery; (**ii**) witness extraction; (**iii**) textual-field transcription; and (**iv**) preservation of relationships among witnesses.
As a conventional OCR measure, we additionally report the *Global Character Error Rate* [30] between the complete ground-truth transcription and the reconciled pipeline transcription, prior to JSON parsing. Table 1 summarizes the metrics used in the study. Where applicable, metrics are reported both overall and only on entries matching in the ground-truth and the pipeline dataset: the latter isolates recognition accuracy among successfully reconstructed entries, whereas the overall results enable the performance to account for missing and spurious entries.

5.2.1 Apparatus structure detection

We evaluate apparatus structure recovery from the one-to-one entry correspondences produced by the alignment procedure described above. An aligned predicted–ground-truth pair is counted as a true positive, an unmatched predicted entry as a false positive, and an unmatched ground-truth entry as a false negative. We then report the proportion of predicted entries successfully aligned with a ground-truth entry (*Entry Precision*), the proportion of ground-truth entries successfully recovered (*Entry Recall*), and their harmonic mean (*Entry* F_1). Results are also broken down by variant type, substitution, omission, transposition, addition, and positive apparatus, to assess whether particular apparatus constructions or editorial signs affect structure recovery.

Table 1: Summary of evaluation metrics and their interpretation.

Name	Interpretation
Baseline OCR	
Global CER	Character error rate of the complete reconciled transcription.
Detection of apparatus structure	
Entry precision	Detected entries that are correct.
Entry recall	True entries that are recovered.
Entry F_1	Harmonic mean of the two.
Witness extraction	
Sigla group exact match	Whole witness group recovered exactly for a given reading.
Sigla precision	Extracted sigla that are correct.
Sigla recall	True sigla that are recovered.
Sigla F_1	Harmonic mean of the two.
Quality of reading transcription	
Reading CER	CER of the transcribed reading.
In-text CER	CER on the in-text portion.
Anchor CER	CER on the anchor (lemma/locus).
Downstream task	
Spearman's coefficient	Measures preservation of relative ranking of pairwise witness similarities.
Adjusted Rand index	Measures similarity of partitions of HAC clustering on the ground truth and the pipeline output.

5.2.2 Witness extraction

Witness extraction is evaluated for sigla appearing both in the `witnesses` fields associated with individual apparatus entries and in the `missing` field, which lists witnesses unavailable on a given page. As the content of the `missing` field is repeated for every entry of a given page, we only count it once per page when computing performance scores. Performance is measured both at the segment level and at the level of individual sigla.
At the segment level, we report the exact-match rate (*Sigla Group Exact Match*), defined as the proportion of ground-truth witness groups for which the predicted set of sigla is identical to the ground-truth set. A group is therefore considered correct only when it contains neither missing nor spurious sigla. At the individual-siglum level, we report the proportion of predicted sigla present in the ground truth (*Sigla Precision*), the proportion of ground-truth sigla recovered by the pipeline (*Sigla Recall*), and their harmonic mean (*Sigla* F_1).

5.2.3 Transcribing textual fields

To measure the capacity of the pipeline to properly transcribe the information related to the textual fields contained within a unit, we compute the CER on the fields `reading` (the value of the reading), `in_text` (the value of the text in the edited text) and `anchor` (the value to anchor the reading to in the case of omissions, additions and some substitutions). Because the reading value can carry significant meaning even with minor changes, such as different spellings of biblical names, we also report the exact-match rate for readings.

5.2.4 Preservation of witness interrelationships

Since the ultimate purpose of the dataset is to enable quantitative analyses of manuscript relationships, we additionally evaluate whether the pipeline preserves patterns of agreement among witness states. We include in the analysis all witness states present in both the ground-truth and pipeline representations, and readings attributed to later hands are treated as distinct states: for example, A^1 is considered separately from A [10]. This representation allows us to assess whether the pipeline preserves distinctions between suc-

cessive hands encoded in the apparatus. The clustering is consequently performed on 126 witnesses.

For each representation, we construct a witness-agreement matrix corresponding to the "grid tables" (*tableaux quadrillés*) used in Quentin's method. Each cell records the number of apparatus entries in which a given pair of witness states supports the same reading. Agreement counts a_{ij} are converted into dissimilarities by max-rescaling and inversion:

$$d_{ij} = 1 - \frac{a_{ij}}{\max_{k \neq l} a_{kl}},$$

with diagonal values set to zero.

To determine whether OCR errors affect downstream quantitative analyses, particularly manuscript clustering, we apply hierarchical agglomerative clustering (HAC) [32] independently to the agreement matrices derived from the ground truth and the pipeline output. Both matrices are clustered using average linkage and a number of clusters varying from $k = 2$ to 5, and the resulting partitions are compared using the Adjusted Rand Index (ARI) [29]. Although hierarchical clustering has previously been used in textual criticism to identify textual families [24], more sophisticated approaches have since been developed [13; 21], but the purpose of this study is, for now, not to make any philological claims concerning relationships among Vulgate witnesses. The goal of reporting these metrics is thus to provide an end-to-end evaluation of whether the OCR-derived data reproduce the global structure obtained from the manually transcribed ground truth.

Finally, we compute Spearman's rank correlation and Pearson's correlation over corresponding agreement counts, considering each distinct pair of witness states once. Spearman's coefficient measures whether the pipeline preserves the relative ordering of pairwise agreements, whereas Pearson's coefficient measures the linear correspondence between their values. Together, these coefficients assess whether the OCR output preserves both the rank-based and numerical structure of the witness-agreement matrix.

5.3 Implementation and data availability

All code developed for this study was written in Python and is available in the companion GitHub repository[3]. The experiments were run on a standalone machine equipped with an integrated GPU and 64 GB of unified memory. Parsing the full 248 pages of Genesis took approximately 80 minutes on this machine. To enable reproducibility, the repository includes both the raw VLM transcriptions and the reconciled transcriptions produced by the full pipeline for the evaluation corpus.

Given the fact that the edition was released in 1926 and has fallen into the public domain, the complete pipeline-generated dataset is released as a *silver-standard* resource under a CC BY 4.0 license. Its expected accuracy is characterized by the evaluation reported in Section 6. We also provide a companion web application for browsing and inspecting the reconstructed critical apparatus.[4]

6 Results and discussion

6.1 Ablation study

Table 2 reports an ablation study comparing the VLM alone (Stage 1), the VLM combined with the dedicated glyph classifier through a naive replacement procedure (Stages 1+2), and the complete OCR pipeline including the specialized reconciliation

[3] The repository URL will be disclosed after anonymous review.

[4] During anonymous review, a screenshot of the application is provided in Appendix Figure 10.

method (Stages 1+2+3). In the naive replacement procedure, candidate notation spans are first extracted from the VLM transcription. The resulting span sequence is then aligned with the classifier-token sequence using a generic sequence-matching procedure based on recursively identified longest contiguous matching subsequences.[5] Classifier predictions associated with matched spans are substituted directly into the VLM transcription, without any of the post-pairing adjudication introduced by the reconciliation method of the full pipeline. The parser constituting Stage 4 was applied identically to all three configurations.

Table 2: Performance of the VLM alone (Stage 1), the VLM and classifier with naive replacement (Stages 1+2), and the full OCR configuration with specialized reconciliation (Stages 1+2+3).

Stages	Evaluation	*Global CER*	*Entry* F_1	*Sigla* F_1	*Text-Field CER*	ρ
1	*All*	61.68	70.32	47.84	47.89	0.37
	Match-only	–	–	70.13	4.90	–
1 + 2	*All*	4.33	95.01	79.05	14.96	0.77
	Match-only	–	–	82.50	4.52	–
1 + 2 + 3	*All*	2.11	96.81	92.68	10.61	0.93
	Match-only	–	–	95.33	2.01	–

Although the VLM produces relatively accurate textual readings once the correct variation unit has been identified (match-only reading CER of 4.90% when using only Stage 1), as expected from a state-of-the-art VLM, it performs poorly on the editorial notation that determines the structure of the apparatus. As a consequence, its entry-level parsing F_1 score reaches only 70.32%, witness extraction is severely degraded ($F_1 = 47.84\%$), and the resulting witness-sharing matrix correlates only weakly with the ground truth (Spearman's $\rho = 0.37$).

Introducing the dedicated classifier through naive replacement produces the largest initial improvement: global CER decreases from 61.68% to 4.33%, entry F_1 increases from 70.32% to 95.01%, and witness-extraction F_1 rises from 47.84% to 79.05%. The corresponding witness-sharing correlation also increases from 0.37 to 0.77. These results confirm that combining a lightweight, domain-specific specialized recognition model with a generic VLM is an efficient solution to increase OCR performance, even with a simpler adjudication phase: although our reconciliation stage relies on edition-specific heuristics, the results of our ablation show that the largest share of OCR improvement lies in the architectural combination of both recognition systems, independently of the particular adjudication strategy employed.

Adding the specialized reconciliation stage further reduces global CER from 4.33% to 2.11% and raises witness-extraction F_1 from 79.05% to 92.68%. Most importantly for the intended downstream application, Spearman's ρ increases from 0.77 to 0.93. The match-only results show the same pattern, with sigla F_1 increasing from 82.50% to 95.33% and reading CER decreasing from 4.52% to 2.01%.

6.2 Apparatus structure recovery

The results regarding the capacity to recover the proper apparatus structure are available in Table 3. Performance by variant type is reported for the classes of textual variation

[5] Using the implementation from Python's `difflib.SequenceMatcher`.

used in the downstream analyses (substitutions, omissions, additions, transpositions, and positive apparatus), while other entry categories, such as editorial notes, are only included in the overall evaluation score.

Table 3: Entry detection precision, recall and F_1 overall and by type of variants.

	Gold count	*Entry precision* (%)	*Entry recall* (%)	*Entry* F_1 (%)
All entries	936	96.40	97.22	96.81
Variant entry type:				
Substitution	681	95.91	96.48	96.19
Omission	84	98.81	98.81	98.81
Addition	41	100.00	100.00	100.00
Transposition	29	100.00	100.00	100.00
Positive apparatus	27	100.00	100.00	100.00

Performance varies slightly according to the type of textual variation. Entries corresponding to *additions*, *transpositions*, and *positive apparatus* are all detected perfectly ($F_1 = 100.00$). As these categories are explicitly introduced by dedicated editorial symbols (respectively, +, ∼, and =), these results demonstrate that our OCR approach properly returns editorial signs.
Omissions reach an F_1 score of 98.81 and *substitutions* an F_1 score of 96.19. The slightly lower performance for substitutions can be explained both by their greater notational variability: unlike the other variant types, substitutions are not always introduced by a unique editorial marker, making their detection more difficult for the parser.
Out of the 681 substitutions present in the gold dataset, 24 were missing from the reconstructed dataset, and 28 were wrongly created.

6.3 Witness extraction

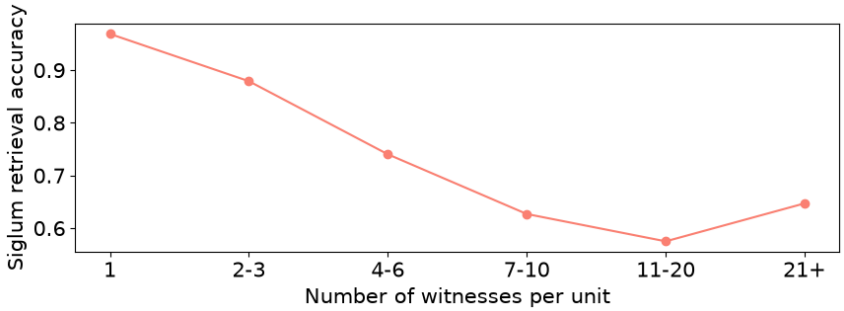
Table 4 reports witness extraction performance, evaluated both over all gold variation units (*All*) and over correctly detected units only (*Match-only*).

Table 4: Evaluation metrics for witness extraction, overall gold units (*All*) and restricted to correctly detected units (*Match-only*).

Metric	**All** (%)	**Match-only** (%)
Sigla precision	92.92	96.13
Sigla Recall	92.44	94.55
Sigla F_1	92.68	95.33
Sigla Group Exact Match	85.90	88.35

Across all units, the pipeline achieves a precision of 92.92%, a recall of 92.44%, and an F_1 score of 92.68%. Restricting the evaluation to correctly detected units increases the F_1 score to 95.33%, showing that a substantial proportion of witness extraction errors originate from missed or incorrectly segmented apparatus entries rather than from the recognition of witness sigla themselves. The exact match rate reaches 85.90% over all units and 88.35% when considering correctly detected units only.
Figure 8 depicts the relationship between witness-group length and the percentage of correctly retrieved sigla within a reading. Witness-group length is defined as the total number of witnesses associated with a reading, including witnesses represented through superscript contractions.

Figure 8: Impact of witness-group length on siglum recall. Bin boundaries were chosen to yield approximately balanced sample sizes across groups.



This figure shows that the proportion of correctly recovered sigla decreases as witness groups become longer, from approximately 96% for individual sigla to below 63% for groups containing more than eleven witnesses, with a slight non-statistically significant raise for the one segment contanining more than 20 different witnesses. This decline suggests that longer witness groups, which tend to involve more complex superscript contractions, are particularly challenging for the pipeline.
This result is particularly relevant for the complete manual verification of the extracted dataset before using it to draw philological conclusions. Witness groups containing a large number of sigla should therefore be prioritized during quality control, since they are associated with substantially lower retrieval accuracy.

6.4 Transcription quality

Table 5 reports OCR performance on the textual fields of the apparatus, evaluated over all gold variation units (*All*) and over correctly detected units only (*Match-only*).

Table 5: OCR performance on textual fields, overall gold units (*All*) and restricted to correctly detected units (*Match-only*).

	All		Match-only	
Text field	**CER** (%)	**Exact** (%)	**CER** (%)	**Exact** (%)
Reading	10.47	91.34	2.26	97.09
In-text	12.81	92.47	0.00	100.00
Anchor	4.12	97.50	4.12	97.50
Overall	10.61	—	2.01	—

A marked improvement is observed when restricting the evaluation to correctly detected units, as the overall character error rate decreases from 10.61% to 2.01%, demonstrating that most transcription errors originate from missed or incorrectly segmented apparatus entries rather than from the recognition of the textual content itself.
This effect is particularly visible for the *reading* and *in-text* fields. The CER of the *reading* field falls from 10.47% to 2.26%, while its exact-match rate increases from 91.34% to 97.09%. Similarly, the *in-text* field is recovered perfectly once the corresponding apparatus entry has been correctly identified (CER = 0.00%, exact match = 100.00%). By contrast, the anchor field has identical performance in both evaluation settings (CER = 4.12%, exact match = 97.50%), indicating that the measured error rate is due to OCR errors

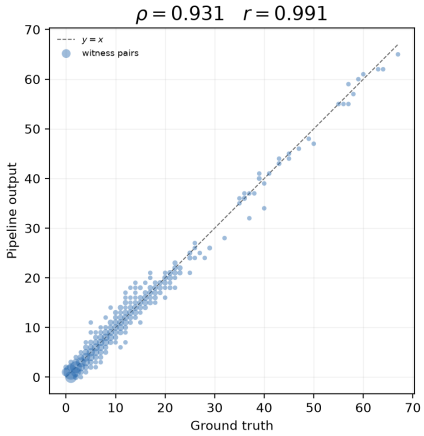
rather than matching ones.

6.5 Preservation of inter-witness relationships

Partitions obtained by hierarchical agglomerative clustering on the ground-truth and pipeline-derived witness-agreement matrices yield ARI values of 1.00 for $k = 2$, 1.00 for $k = 3$, 0.75 for $k = 4$, and 0.93 for $k = 5$. These results indicate that the broad partition structure is preserved exactly at larger clustering resolutions, while finer subdivisions are more sensitive to differences introduced by the extraction pipeline. The lower agreement observed at $k = 4$ can be epxlained by the instability of this particular cut: the fourth cluster contains only two witnesses, making its assignment especially sensitive to small changes in the dissimilarity matrix. The results therefore suggest that the pipeline reliably preserves the broad structure of the agreement data under this downstream method, while specific fine-grained subgroup assignments should be confirmed on the gold-standard dataset.

In addition, the Pearson and Spearman correlation coefficients between the corresponding entries of the two agreement matrices are 0.991 and 0.931, respectively. The very high Pearson correlation indicates that pairwise agreement counts retain an almost perfectly linear relationship, while the high Spearman correlation shows that the relative ranking of witness similarities is also largely preserved. Figure 9 illustrates this result by plotting pairwise witness-agreement counts derived from the pipeline output against those obtained from the ground truth.

Figure 9: Comparison of pairwise witness-agreement counts derived from the ground truth and the pipeline output.



Indeed, most observations lie close to the identity line ($x = y$), indicating that the agreement counts themselves are preserved with only limited deviations. Notably, deviations from the identity line are not concentrated on particular witnesses or members of a single textual family, and OCR errors therefore appear to behave as unsystematic noise with respect to the downstream analysis.

Taken together, these results show that the silver-standard dataset can therefore already support exploratory analyses of broad witness affinities, although these results should be taken with caution given the simplicity of the clustering method. Any specific philological conclusions should consequently be validated against the edition itself, or performed on the gold-standard dataset when released.

7 Limits of the study

We identify four limitations of the current study, which will be addressed in future work.

Absence of linkage with the edited text The current dataset does not yet link apparatus readings to the edited text in substitution entries for which the adopted reading is not explicitly repeated in the apparatus. Short-term work will therefore focus on linking these entries to the running text by identifying the candidate span that minimizes the normalized Levenshtein distance to the apparatus lemma, following the approach proposed by Bambaci et al. [3].

Silver-standard status of the dataset The dataset and corresponding web application are currently released as a *silver-standard* resource, whose estimated reliability follows the numbers presented in Section 6. The complete apparatus of Genesis will be manually corrected in the coming months to produce a gold-standard dataset suitable for philological analysis, a process considerably accelerated by the accuracy of the extracted data. Corrected chapters will be released and studied incrementally, since each chapter already constitutes a meaningful unit for philological analysis.

Absence of an end-to-end fine-tuned baseline The study does not compare the modular pipeline with an end-to-end model trained to generate structured records directly from page images, since creating the required training corpus would entail manually structuring a substantial part of the apparatus. Future work will investigate whether the progressively corrected Genesis dataset can support an end-to-end baseline and whether such a model improves performance or transferability to other editions.

Specificity of the pipeline to the Benedictine Vulgate Although the general workflow can be adapted to other critical editions, its current implementation is specifically tailored to the typographical and editorial conventions of the 18-volume Benedictine Vulgate. Adapting it to another edition would require retraining the lightweight glyph classifier, modifying the reconciliation rules, and redefining the formal grammar according to the structure and notation of the target apparatus. However, the specificity of our approach is mitigated by the ablation study, which shows that a more generic reconciliation strategy already yields substantial improvements over the VLM alone. This suggests that the proposed two-model architecture could be transferred to other critical editions with similarly complex sigla systems.
Additionally, within the edition itself, we acknowledge that the evaluation corpus is contiguous and limited to Genesis 1–4: although it contains all constructions, the reported accuracy may not capture variation in print quality or apparatus complexity across the complete editions.

8 Conclusion

This study demonstrates that combining a VLM, specialized classifier, rule-based reconciliation, and formal-language parser can reliably transform the Benedictine Vulgate apparatus into structured data while globally preserving the witness-agreement patterns used in downstream analysis.
Our immediate work will focus on manually correcting the complete dataset for Genesis and using it to investigate the classification and stemmatic relationships of the Vulgate witnesses. We will then progressively extend the dataset to other biblical books, enabling

broader investigations into the transmission of the Latin Vulgate. Further methodological work will focus on linking substitution readings to their corresponding lemmas in the edited text to reconstruct the readings of the individual witnesses.

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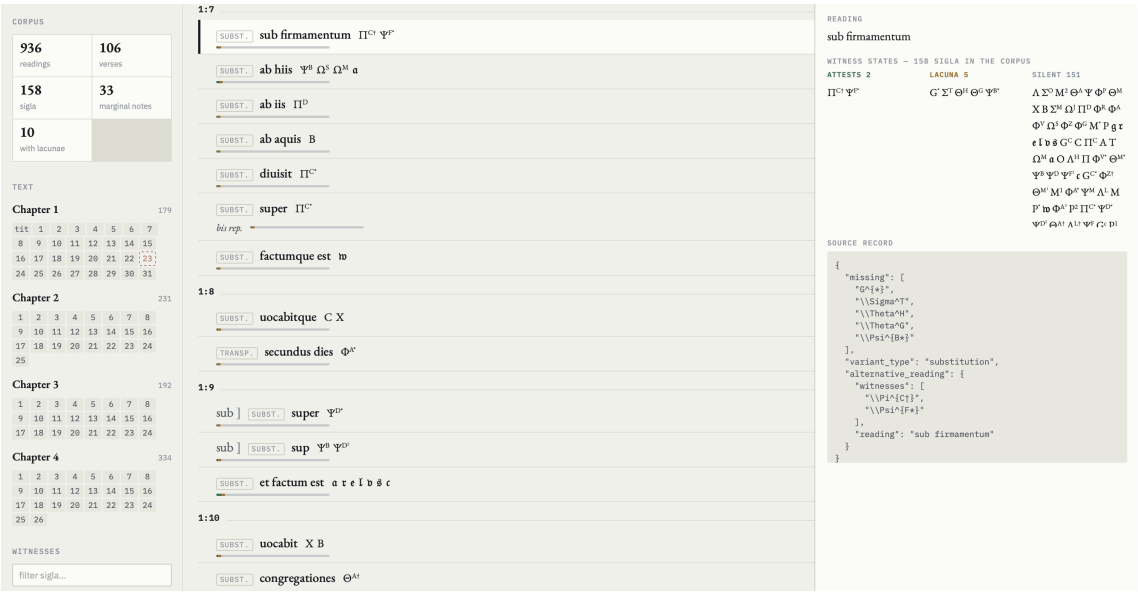
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Appendix

A. Screenshot of companion Web application

Figure 10: Screenshot of companion Web application



B. Reconciliation procedure

The reconciliation stage combines the VLM transcription of each complete apparatus region with the ordered tokens, bounding boxes, and confidence scores produced by the specialized classifier. The VLM output is used as the carrier text, while the classifier is used to correct or recover sigla, Fraktur sequences, and editorial signs. The procedure is deliberately conservative: when the two streams cannot be paired reliably, the VLM reading is retained and the case is reported for review. All thresholds were fixed on development pages outside the evaluation chapters and should be understood as edition-specific reconciliation parameters. The implementation of this reconciliation procedure is fully available in the companion GitHub repository.

Candidate preparation The VLM transcription is first normalized by removing formatting artifacts and splitting composite superscript groups into atomic units. Candidate spans are then extracted from their LaTeX, HTML, or Unicode representations. Each is assigned an alignment key: the base character for a manuscript siglum, the symbol itself for an editorial sign, and the recognized letter or a generic key for a Fraktur cluster. Plain Latin capitals are treated as sigla only when they carry a modifier, limiting confusion with ordinary text. The positive-apparatus sign = is not extracted because it has no classifier class and it is retained from the VLM to be interpreted by the downstream grammar. Classifier tokens are also placed in line-major reading order.

Global sequence alignment The two token sequences are aligned by minimum-cost Needleman–Wunsch. The costs are given in Table 6, where c_j is the classifier family confidence for token t_j.

Table 6: Costs used to align VLM spans and classifier tokens.

Operation	Cost
Pair identical alignment keys	0
Pair different manuscript-siglum bases	0.45
Leave a VLM span unmatched	0.35
Leave a classifier token unmatched	$0.25 + 0.5c_j$

A cross-key pairing is allowed only between siglum-shaped candidates when the classifier's family and base confidences are both at least 0.90. This lets the classifier correct a VLM base error without allowing unrelated symbols to pair solely because they occupy similar sequence positions.
The VLM does not provide character-level coordinates. We therefore obtain an approximate location for each VLM word by aligning it with ink components classified as background and interpolating the location of symbol spans within the word. Geometry does not enter the dynamic-programming cost, since an incorrect positional estimate could otherwise displace a longer correct alignment. It is used only after alignment to recover a compatible unmatched span-token pair when both occupy the same patch of ink.

Post-alignment adjudication An aligned pair is not automatically written to the output. First, when the bases agree but the modifier sets or digit/asterisk signatures conflict, the pair is normally rejected because repeated base sigla may have caused an occurrence shift. The classifier is retained only when:

1. the VLM modifier is illegal for that base and the classifier proposes a legal, confident form;
2. the difference belongs to a recurrent, directional VLM confusion observed during development and the modifier confidence is at least 0.99; or
3. both readings locate the same ink and the classifier's family, base, and modifier confidences are all at least 0.90.

Second, a generic Fraktur pair is accepted only if one decoded letter sequence is a subsequence of the other or their sequence similarity is at least 0.50. Third, when the classifier output is a strict subsequence of a longer VLM siglum that is well formed according to the edition's siglum inventory, the VLM is retained. Replacement or splitting is permitted only when the additional VLM modifiers are illegal or geometry shows that a neighboring full-height witness was absorbed into the superscript group. Finally, immediately duplicated modifiers or duplicated plain-text copies of a corrected siglum are collapsed when their normalized forms agree.

Recovery of unmarked material Unmatched classifier tokens are normally not inserted into the VLM transcription and are reported for review. Two constrained exceptions address recurrent VLM omissions.
Consecutive unmatched Fraktur tokens are grouped, and the interval between their nearest aligned neighbors is searched for a compatible lowercase VLM word. When a match is found, the VLM letters are retained and only their typography is changed; an individual classifier letter replaces the VLM only at confidence ≥ 0.99. If the complete run is absent,

it may be inserted only when it contains at least three glyphs, all with confidence ≥ 0.99, and lies on the same line within 120 pixels of an aligned token.
An omitted vertical separator may likewise be inserted only at confidence ≥ 0.99, with the expected tall and narrow geometry, and within 150 pixels after an aligned token on the same line. No unanchored insertion is performed for other editorial signs.
All accepted decisions are finally rendered from a common intermediate representation into canonical LaTeX and human-readable Unicode. The report records corrections, excluded predictions, unmatched items, and all performed recoveries.